# MIL-M-8916A(ASG)
## 2 MAY 1969

Superseding
MIL-M-8916(ASG)
28 March 1962

MILITARY SPECIFICATION

MAGNESIUM ALLOY BAR, ROD, AND
SPECIAL SHAPED SECTIONS, EXTRUDED (HM31A-T5)

This specification has been approved by the Department of the Air Force and by the Naval Air Systems Command.

1. SCOPE

* 1.1 This specification covers HM31A magnesium alloy bar, rod, and special shaped sections in the artificially aged (T5) temper.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-M-3171 | Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on |

STANDARDS

Federal

| | |
|---|---|
| FED. TEST METHOD STD. NO. 151 | Metals; Test Methods |
| FED. STD. NO. 184 | Identification Marking of Aluminum, Magnesium and Titanium |
| FED. STD. NO. 245 | Tolerances for Aluminum Alloy and Magnesium Alloy Wrought Products |

Military

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-649 | Aluminum and Magnesium Products, Preparation for Shipment and Storage |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9530

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* 2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

| | |
|---|---|
| ASTM E8 - 66 | Tension Testing of Metallic Materials |
| ASTM E21 - 66T | Short Time Elevated Temperature Tension Tests of Materials |

(Copies of ASTM Standards may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Atomic Energy Commission

Code of Federal Regulations - Title 10, Part 20 - Standards for Protection Against Radiation

(Application for copies should be addressed to the Division of Licensing and Regulation, U.S. Atomic Energy Commission, Washington, D.C. 20545.)

3. REQUIREMENTS

3.1 Material. - The material shall be such as to produce bar, rod, and shapes conforming to this specification.

* 3.2 Chemical composition. - The chemical composition shall be as specified in table I, when tested in accordance with 4.4.2.

TABLE I. Chemical composition

| Element | Percent by weight | |
|---|---|---|
| | Min. | Max. |
| Thorium | 2.5 | 3.5 |
| Manganese | 1.2 | - |
| Other impurities, each | - | 1/ 0.10 |
| Other impurities, total | - | 1/ 0.30 |
| Magnesium | Remainder | |

1/ Analysis shall regularly be made only for the elements specifically mentioned in table I. If, however, the presence of other elements is suspected or indicated in amounts greater than the specified limits, further analysis shall be made to determine that these elements are not present in excess of the specified limits.

3.2.1 The contractor shall furnish an analysis of each melt represented in the lot showing the percentage of each of the elements specified in table I.



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3.2.2 Chemical analysis by the contractor of the individual melts may be waived at the discretion of the procuring activity, provided that the producer's method of composition control is acceptable or that all material in the lot can be identified as being from melts previously analyzed and found to be in conformance with the chemical composition requirements.

* 3.3 Mechanical properties. - When tested as specified in 4.5.2, the mechanical properties of the extruded bar, rod, and shapes shall be as specified in table II.

* TABLE II. Mechanical properties (HM31A-T5)

| Form | Specified cross-sectional area (sq. in.) | Tensile strength, minimum (psi) | Yield strength at 0.2 percent offset minimum (psi) | Elongation, percent in 2 inches, or 4D minimum |
|---|---|---|---|---|
| Bars, rods, and shapes (solid) | Under 4.000 | 37,000 | 26,000 | 4 |

3.3.1 Sizes not specifically covered. - Mechanical properties for sizes of material not covered by this specification shall be as specified in the contract or order (see 6.2).

* 3.3.2 Tensile properties at 600° F. - When tested as specified in 4.5.3.2, material under 4.000 square inches in area shall be capable of conforming to the minimum mechanical properties specified in table III at 600° F.

TABLE III. Minimum mechanical properties at 600° F (HM31A-T5)

| Tensile strength, minimum (psi) | Elongation, percent in 2 inches, minimum |
|---|---|
| 12,000 | 10 |

* 3.4 Dimensional tolerances. - The extruded bar, rod, and shapes shall not vary from the specified dimensions, straightness, contour, twist, surface roughness, flatness, leg angles and corner and fillet radii by amounts greater than the applicable tolerances specified in FED. STD. NO. 245 for cross-sectional dimensions, straightness,



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* contour, twist, and surface roughness of extruded bar, rod, and shapes; for squareness of cut and for extruded and rolled bar, rod, and shapes; for flatness and corner and fillet radii of extruded bar and shapes; and for leg angle of extruded shapes.

3.5 Marking for identification. - Unless otherwise specified in the contract or order (see 6.2), marking shall be in accordance with FED. STD. NO. 184. In addition to the commercial alloy designation, the number of this specification shall be applied to each extrusion.

3.6 Precautionary information. - The processing, handling, and shipment of products covered by this specification shall be as specified in the Code of Federal Regulations, Title 10, Part 20, titled "Standards for Protection Against Radiation" (see 6.2).

3.7 Protective treatment. - The bar, rod, and shapes, shall be either oiled or chrome-pickled, as specified (see 6.2), as a final process before shipment. Unless otherwise specified (see 6.2), the extrusions shall be given a chrome-pickle treatment in accordance with type I of MIL-M-3171.

3.8 Workmanship. - The extruded bar, rod, and shapes shall be uniform in quality and condition; clean, sound, smooth, and free from seams, slivers, laminations, grooves, cracks, and other injurious defects. Discoloration caused by heat treatment will not be cause for rejection.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. - A lot shall consist of material of the same section, size, and temper, manufactured during the same production period.

4.3 Examination. - Each bar, rod, and shape shall be carefully examined to determine conformance with this specification with respect to workmanship. Sufficient spot checks shall be made to insure conformance with the tolerances specified. On approval by the procuring activity, a system of statistical quality control may be used for dimensional and workmanship examination.



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4.4 Chemical analysis. -

4.4.1 Sampling. - A sample for chemical analysis shall be obtained from each 4,000 pounds or less of material comprising the lot. When the sample for analysis is taken from the fabricatedd product, it shall consist of equal quantities of chips taken from at least four representative pieces of the full thickness of the material.

4.4.2 Method of analysis. - The sample selected shall be analyzed by the wet chemical method in accordance with Method 111 of Federal Test Method Standard No. 151, or spectrographic method in accordance with Method 112 of Federal Test Method Standard No. 151 to determine conformance with 3.2. A single analysis of a composite sample may be made. In case of dispute, the analysis by the wet method (Method 111) shall be the basis for acceptance.

4.5 Mechanical properties (tension test). -

* 4.5.1 Sampling. - From material having a nominal weight of less than 1 pound per lineal foot, one tension test sample shall be selected from each lot for each 1,000 pounds, or fraction thereof. From material having a nominal weight of 1 pound or more per lineal foot, one tension test sample shall be selected from each lot for each 1,000 feet, or fraction thereof.

* 4.5.2 Location of test specimens. - Tension test specimens shall be taken parallel to the direction of extrusion. For material 1-1/2 inch and less in diameter or thickness, when not tested in full section, the specimen shall be taken from the center of the section. For material greater than 1-1/2 inch in diameter or thickness, the specimen shall be taken midway between the center and the surface. For extruded shapes, the specimen shall be taken from the predominant part of the shape and the location shall be midway between the center and the surface.

* 4.5.3 Test specimens. - Tension test specimens shall conform to ASTM E8. Whenever applicable, the material may be tested in full section. For material less than 1/2 inch in section thickness and of suitable width, which is not tested in full section, a rectangular type of specimen shall be used. For material 1/2 inch or more in section thickness, which is not tested in full section, a round tension test specimen shall be used. For shapes from which a standard test specimen cannot be taken, and which cannot be tested satisfactorily in full section, a strip taken from the section shall be tested.

4.5.4 Tension tests. -

* 4.5.4.1 Room temperature. - The test shall be performed in accordance with ASTM E8.

* 4.5.4.2 Elevated temperature. - The test shall be performed in accordance with ASTM E21. Unless otherwise specified (see 6.2) test temperature shall be 600° F.



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* 4.5.5 Yield strength. - Yield strength shall be determined by the offset method in accordance with ASTM E8.

4.6 Rejection and retests. -

4.6.1 Rejection. - If any individual test specimen fails to conform to this specification, the lot represented by the specimen shall be rejected, except when a statistical sampling plan is used.

4.6.2 Retests. - If a lot of material is rejected in accordance with 4.6.1, retests of that lot are permitted in accordance with the provisions of Federal Test Method Standard No. 151 by (a) taking twice the original number of test specimens from an area in the original sample adjacent to the area represented by the failure, or (b) taking the original number of test specimens from an area in each of two other extruded bars, rods, or shapes identical to the area in the original sample represented by the failure. Should any of these specimens fail to conform to this specification, the entire lot shall be rejected.

4.7 Packaging, packing, and marking. - Preparation for delivery shall be examined for conformance to section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing (see 3.6). -

5.1.1 Unless otherwise specified in the contract or order, all material shall be separated by temper, size, and thickness and packed in crates or boxes of the size commonly used for shipment. When specified (see 6.2), individual extruded bar, rod, and special shaped sections which have been given a chrome pickle surface treatment shall be separated by a noncorrosive, nonhygroscopic paper to protect the finish. Oiled bar, rod, or special shaped sections need not be separated by paper.

* 5.1.2 Levels of packing. - All bar, rod, and special shaped and extruded sections shall be packaged and packed in accordance with MIL-STD-649 (see 6.2). The procuring activity will specify the levels required. When no level is specified, level C shall be used.

5.2 Marking. -

5.2.1 In addition to any special marking required by the contract or order (see 6.2), marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - HM31A alloy is used primarily for components requiring weldability and good strength-to-weight ratio up to 600° F.



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6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Length, size (die number and shapes).
(c) Mechanical properties for sizes not covered by table II (see 3.3.1).
(d) Tolerances for material having dimensions outside the specification limits.
(e) Purpose for which bar, rod, or special shapes, extruded, is intended.
(f) Precautionary information (and Atomic Energy Commission License number)(see 3.6).
* (g) Protective treatment (see 3.7).
(h) Elevated temperature for tension test if other than as specified in 4.5.4.2.
(i) Interleave packaging, if required (see 5.1.1).
(j) Level of packaging and packing required (see 5.1.2).
(k) Special marking, if required (see 3.5 and 5.2.1).

* 6.3 Marginal indicia. - The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - AS
Air Force - 11

Preparing activity:
Air Force - 11

Project No. 9530-0134



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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-M-8916A(ASG) MAGNESIUM ALLOY BAR, ROD, AND SPECIAL SHAPED SECTIONS, EXTRUDED (HM31A-T5)

| ORGANIZATION | CITY AND STATE |
|---|---|

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $ |
|---|---|---|

MATERIAL PROCURED UNDER A

( ) Direct Government Contract ( ) Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE?

   ( ) YES ( ) NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

DD Form 1426

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